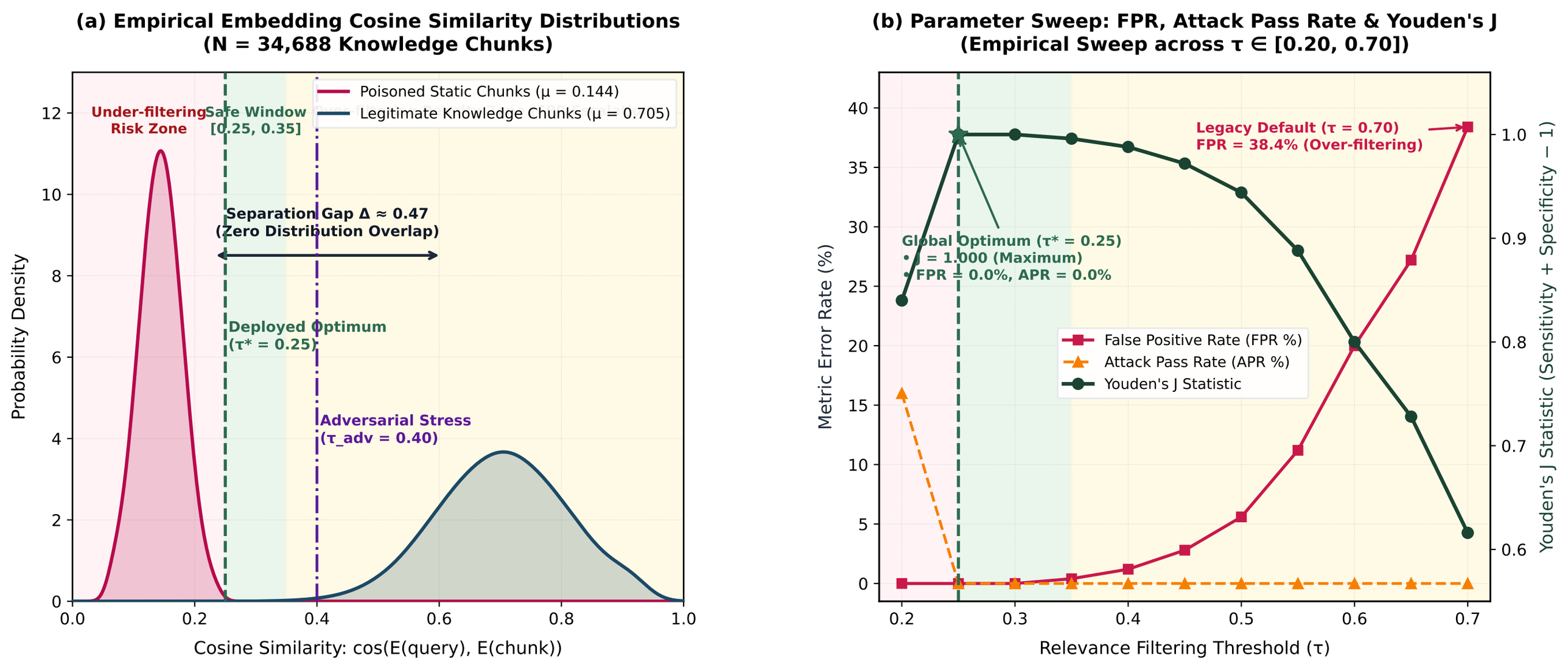

(a) Empirical Embedding Cosine Similarity Distributions
(N = 34,688 Knowledge Chunks)
Poisoned Static Chunks (μ = 0.144)
Legitimate Knowledge Chunks (μ = 0.705)
Under-filtering
Risk Zone
Safe Window
[0.25, 0.35]
Separation Gap Δ ≈ 0.47
(Zero Distribution Overlap)
Deployed Optimum
(τ* = 0.25)
Adversarial Stress
(τ_adv = 0.40)
Probability Density
Cosine Similarity: cos(E(query), E(chunk))
(b) Parameter Sweep: FPR, Attack Pass Rate & Youden's J
(Empirical Sweep across τ ∈ [0.20, 0.70])
Legacy Default (τ = 0.70)
FPR = 38.4% (Over-filtering)
Global Optimum (τ* = 0.25)
• J = 1.000 (Maximum)
• FPR = 0.0%, APR = 0.0%
False Positive Rate (FPR %)
Attack Pass Rate (APR %)
Youden's J Statistic
Metric Error Rate (%)
Youden's J Statistic (Sensitivity + Specificity − 1)
Relevance Filtering Threshold (τ)